

Q3) Difference between if-else and switch-case and their use in IoT

Both if-else and switch-case are **decision-making statements** used to control the flow of a program.

Difference between if-else and switch-case

if-else	switch-case
It is used to check multiple conditions and ranges .	It is used to check one variable against multiple fixed values .
It can compare using operators like <, >, <=, >=, ==.	It mainly checks equality (==) with constant values.
It works with different variables in different conditions.	It works with a single variable/expression.
Execution is slower when there are many conditions.	Faster and cleaner for many fixed choices.
No break statement is required.	break is used to stop execution after a matching case.
Best for sensor value comparisons.	Best for selecting modes/states.

Syntax of if-else

```
if(condition)
{
    // code if condition is true
}
else
{
    // code if condition is false
}
```

Example:

```
int temperature = 40;

if(temperature > 35)
{
```

```
    fanON();
}
else
{
    fanOFF();
}
```

Syntax of switch-case

```
switch(variable)
{
    case value1:
        // code
        break;

    case value2:
        // code
        break;

    default:
        // code
}
```

Example:

```
int mode = 2;

switch(mode)
{
    case 1:
        Serial.println("Automatic Mode");
        break;

    case 2:
        Serial.println("Manual Mode");
        break;

    case 3:
        Serial.println("Sleep Mode");
        break;

    default:
        Serial.println("Invalid Mode");
}
```

Uses in IoT

1) if-else applications:

Used when IoT devices need to take decisions based on **sensor values**.

Examples:

Smart Street Light

```
if(light < 300)
{
    LED_ON();
}
else
{
    LED_OFF();
}
```

Temperature Monitoring

```
if(temp > 40)
{
    Fan_ON();
}
else
{
    Fan_OFF();
}
```

Smart Irrigation

```
if(soilMoisture < 500)
{
    Pump_ON();
}
else
{
    Pump_OFF();
}
```

2) switch-case applications:

Used when IoT devices have **different modes or states**.

Examples:

Smart Home Modes:

Mode 1 → Normal Mode

Mode 2 → Security Mode

Mode 3 → Power Saving Mode

Mode 4 → Maintenance Mode

Code:

```
switch(mode)
{
  case 1:
    normalMode();
    break;

  case 2:
    securityMode();
    break;

  case 3:
    powerSavingMode();
    break;
}
```

Examples:

- LDR, temperature, humidity → if-else
- Menu, device state, operating mode → switch-case

Q4) What are System States and how to manage them?

System States

A **system state** represents the **current condition or mode of operation of a system at a particular time.**

In IoT, a device can work in different modes depending on:

- sensor values
- user input
- commands

- time
- events

Each mode is called a **state**.

Example: Smart Street Light System

A street light can have different states:

OFF State

|

| (Darkness detected)

↓

ON State

|

| (Daylight detected)

↓

OFF State

Here:

- **OFF state** → LED is OFF during daytime
- **ON state** → LED is ON during night

Example: Smart Home System

Different states:

1. **Normal State**
 - Devices operate normally
2. **Security State**
 - Motion sensors active

- Alarm system ON
- 3. **Power Saving State**
 - Unnecessary devices OFF
- 4. **Emergency State**
 - Alarm activated

How to Manage System States?

System states are managed using a **Finite State Machine (FSM)**.

FSM has:

1. **States** → Different operating modes
2. **Events/Inputs** → Conditions causing change
3. **Transitions** → Moving from one state to another

Methods to manage states in programming:

1) Using variables

Store the current state:

```
int state = 1;
```

Example:

state = 1 → OFF mode

state = 2 → ON mode

state = 3 → Sleep mode

2) Using if-else

```
if(state == 1)
```

```
{
```

```
    LED_OFF();
```

```
}
```

```
else if(state == 2)
```

```
{
```

```
    LED_ON();
```

```
}
```

3) Using switch-case (preferred)

```
switch(state)
```

```
{
```

```
    case 1:
```

```
        LED_OFF();
```

```
        break;
```

```
    case 2:
```

```
        LED_ON();
```

```
        break;
```

```
    case 3:
```

```
        sleepMode();
```

```
        break;
```

```
}
```

switch-case is commonly used because states usually have **fixed values**.

IoT Applications of System States:

IoT Device	Possible States
Smart light	ON, OFF, Auto
Smart fan	OFF, Low, Medium, High
Security system	Normal, Alert, Alarm
Smart irrigation	Idle, Checking, Watering
Robot	Stop, Forward, Reverse